Title: Understanding Pidgin English Through NLP: Translation, Sentiment, and Social Media Discourse

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Abstract

Spoken by millions across West Africa, especially in Nigeria, **Nigerian Pidgin English** plays a vital role in everyday communication, cultural identity, and online expression. However, it remains largely excluded from mainstream language technologies such as translation tools, chatbots, and sentiment analyzers. This research project addresses that gap by collecting real-world Pidgin text from platforms like Twitter, Reddit, and Facebook, and creating a parallel corpus with English translations. These text pairs will be used to fine-tune existing multilingual translation models and improve machine understanding of Pidgin.

In addition to translation, the project focuses on **sentiment analysis** in Pidgin, aiming to capture the emotional tone—such as humor, frustration, or support—embedded in social media posts. Due to Pidgin's informal grammar and regional slang, standard sentiment tools often fail to accurately interpret meaning. To overcome this, we will develop tailored NLP models and lexicons that reflect the unique linguistic structures of Pidgin. The analysis will offer insights into how emotion and expression are shaped within digital discourse in Nigerian communities.

As a practical demonstration, we will also build a **lightweight web application** using React and FastAPI. This tool will allow users to input text in either Pidgin or English, receive real-time translations, view sentiment labels (positive, neutral, negative), and eventually explore cultural idioms through interactive tooltips. It will serve both non-Pidgin speakers looking to access local digital content and Pidgin speakers seeking validation of their language in modern technology.

Beyond technical development, this project emphasizes the cultural value of Pidgin as a resilient, expressive language often overlooked in academic and digital contexts. By enhancing technological support for Pidgin, we aim to foster digital inclusion and linguistic equity for millions of speakers. Ultimately, this work builds on efforts like NaijaSenti, JW300, and Masakhane to promote AI tools that are more inclusive, culturally aware, and useful to African communities.

Introduction

Pidgin English, including regional varieties such as Nigerian Pidgin, Cameroonian Pidgin, Tok Pisin (spoken in Papua New Guinea), and Solomon Islands Pijin, serves as a vital **lingua franca** across many multilingual societies in Africa, the Caribbean, and the Pacific. These languages emerged from historical contact through trade, colonization, and cultural exchange, enabling communication between people who do not share a common native language. Despite their widespread use in everyday life, Pidgin languages remain **marginalized** in formal education systems, mainstream media, and most notably, in modern technological tools.

With the rise of **Natural Language Processing (NLP)**, there is a growing opportunity to bridge this gap. Developing NLP tools specifically designed for Pidgin languages can significantly improve **machine translation**, **sentiment analysis**, and the ability to interpret **social media discourse**, a domain where Pidgin is widely used for cultural expression, political commentary, and community engagement.

While some progress has been made, such as the use of transformer-based models trained on datasets like **AfriBERTa**, NLP for Pidgin remains a **low-resource field**, with limited data and tools tailored to its unique structure and informal tone.

This research seeks to fill that gap by advancing NLP applications for **Nigerian Pidgin English**, focusing on translation, sentiment classification, and analysis of online discourse. In doing so, it aims not only to improve language technology but also to promote **digital inclusion**, support **cultural preservation**, and empower communities whose voices are often excluded from global AI systems.

Problem Statement

Pidgin English is a widely spoken lingua franca across several West African nations, particularly Nigeria, where it serves as a vital medium of everyday communication, cultural expression, and online discourse. Despite its prominence and rich sociolinguistic significance, Pidgin English remains largely underrepresented in Natural Language Processing (NLP) research and tools. Existing NLP models and translation systems are primarily optimized for standard global languages and often fail to capture Pidgin English's unique grammar, vocabulary, and contextual nuances. This linguistic gap presents significant challenges for automatic translation, sentiment analysis, and understanding social media discourse in contexts where Pidgin is prevalent. The lack of computational resources and datasets further hinders academic and technological advancements. Consequently, existing NLP tools often misinterpret or ignore the nuances of Pidgin, leading to inaccurate translations, misclassified sentiments, and limited cultural insights from social media content. Addressing this gap is crucial not only for improving language technologies but also for ensuring digital inclusion, cultural preservation, and equitable representation in AI-driven communication systems.

Objectives

- Objective 1: Translating pidgin to English and finding and incorporating modern pidgin
- Objective 2: Sentiments; classifying it as a positive, negative, or neutral pidgin
- Objective 3: We want pidgin English to be recognized
- Objective 4: Research question or hypothesis, if applicable

Literature Review

Natural Language Processing (NLP) for low-resource languages has gained momentum in recent years, yet Pidgin English remains significantly underrepresented. A few foundational efforts provide groundwork for this study.

The **NaijaSenti** corpus (Muhammad et al., 2022) is a large-scale, manually annotated dataset focused on sentiment analysis in Nigerian languages, including Pidgin. With approximately 14,000 annotated Pidgin tweets, it enables improved sentiment modeling, especially when paired with adapted sentiment lexicons like the 300-token VADER enrichment developed by Oyewusi et al. (2020). While effective for emotion classification, these datasets are not designed for translation.

For machine translation, the **Masakhane NMT project** has been instrumental in pioneering African language modeling using neural networks. Their work includes experiments with multilingual transformers such as **mBART** and **mT5** on Pidgin, but the available parallel corpora remains limited in both size and domain variety.

The **JW300 corpus** offers aligned religious text pairs between Pidgin and English, largely derived from Bible translations. While useful for structure and syntax alignment, its formal and spiritual tone differs drastically from casual, internet-based Pidgin used in everyday discourse.

Additionally, the **AfriBERTa** and **NLLB-200** models demonstrate that large-scale multilingual pretraining can include underrepresented languages like Pidgin, but such models are rarely fine-tuned on domain-specific content like social media posts, where most real-world Pidgin is now used.

This research distinguishes itself by targeting modern, real-world Pidgin, particularly from social media platforms, to construct a custom parallel corpus and fine-tune existing NMT models for improved bidirectional translation. In doing so, it fills a critical gap between available structured datasets and informal online language use, while opening pathways for further discourse analysis and application development.

Methodology

Research Design:

This study adopts a mixed-methods approach, combining computational techniques (machine translation modeling and evaluation) with sociolinguistic analysis (discourse patterns in online Pidgin).

Data Annotation:

After scraping posts, a team of bilingual annotators will manually translate selected Pidgin texts into English. Posts will be tagged with sentiment labels (positive, negative, neutral) using dual-review to ensure reliability, with adjudication for disagreements.

Platform & Tools:

- Twitter via Tweepy, Reddit via PRAW, and Facebook via Graph API will be used for scraping.
- Preprocessing includes language filtering (using FastText), emoji/text normalization, and removal of irrelevant posts.
- Translation modeling will rely on fine-tuning **mBART**, **mT5**, and **NLLB-200**, using Hugging Face's Transformers and PyTorch Lightning.

Supplementary Datasets:

- **NaijaSenti** – for sentiment comparison and optional model enrichment
- **JW300** – as a baseline parallel corpus
- **Masakhane NMT** – for model architecture and training reference

Tools and Techniques:

- **Scraping and preprocessing**: Python (Tweepy, PRAW, Facebook Graph API), language detection (FastText), text normalization
- **Modeling**: Fine-tuning of existing multilingual models (e.g., **mBART**, **mT5**, **NLLB-200**), trained on the custom parallel corpus
- **Evaluation Metrics**: Translation models will be assessed using **BLEU**, **METEOR**, and **chrF** scores, as well as human evaluations by bilingual speakers. Sentiment models will be evaluated with **F1 score**, **precision**, and **recall**, using a 70/15/15 train/dev/test split. Annotation inter-rater agreement will be tracked using **Cohen's Kappa**.
- **Web App Tool**: We will develop a simple bilingual demo web app using **React** (frontend) and **FastAPI** (backend). The app will allow users to input Pidgin or English text and view the translated output in real-time. The primary goal is accessibility: enabling non-Pidgin speakers to understand informal digital content and vice versa. Core functionality will include text input, instant translation using our fine-tuned model, and confidence scoring. Future extensions may include sentiment tagging and examples of cultural idioms.

Qualitative Analysis:

A thematic review of Pidgin-English posts will be conducted to identify how language is used for:

- Cultural identity
- Humor and memes
- Political or social commentary

This will help contextualize the NLP outputs and reveal deeper insights about the digital role of Pidgin.

Expected Outcomes

1. **Development of Pidgin-English NLP Tools:**
   Creation or adaptation of Natural Language Processing models specifically trained to process, understand, and generate Nigerian Pidgin English, including tools for translation, sentiment analysis, and text classification.
2. **Improved Machine Translation Accuracy:**
   Enhanced bilingual (Pidgin–English) translation systems that better handle the grammatical structures, idioms, and contextual meanings unique to Pidgin English.
3. **Sentiment Analysis Framework for Pidgin:**
   A robust sentiment analysis model capable of accurately interpreting emotional tones and opinions expressed in Pidgin across social media platforms.
4. **Annotated Pidgin-English Dataset:**
   Compilation and release of a curated, linguistically diverse dataset of Pidgin English text from social media or other digital platforms, annotated for translation and sentiment.
5. **Insights into Online Social Discourse:**
   Analysis of how Pidgin English is used in online conversations to convey cultural identity, political opinions, humor, resistance, or solidarity, offering sociolinguistic insights.

6. **Contribution to Inclusive AI:**
   Advancement in ethical and inclusive AI by expanding NLP capabilities to underrepresented languages and dialects, aligning with global efforts to democratize language technologies.

Evaluation Plan and Timeline

July 2025 – Data Collection and Preparation

- Scrape Pidgin-English text pairs from social media platforms (Twitter, Reddit, Facebook)
- Clean and normalize text for consistent tokenization
- Begin manual annotation of translation and sentiment labels

August 2025 – Dataset Finalization and Initial Modeling

- Finalize annotated dataset for translation and sentiment tasks
- Fine-tune multilingual models (e.g., mBART, mT5) on custom parallel corpus
- Evaluate translation accuracy using BLEU and other metrics

September 2025 – Sentiment Model Development and Discourse Analysis

- Train sentiment analysis models using NaijaSenti and enriched lexicons
- Evaluate performance using F1 score and confusion matrices
- Begin qualitative analysis of Pidgin in online discourse (identity, humor, protest, etc.)

October 2025 – Tool Integration and Application Prototyping

- Build lightweight NLP tools (translation, sentiment, tagging) for browser or mobile use
- Develop simple web interface for live text input and model predictions
- Integrate Pidgin NLP tools with social media monitoring or education platforms

November 2025 – Final Evaluation and Report Writing

- Conduct end-to-end evaluation of system performance on real-world test data
- Refine models and documentation for release
- Write and finalize project report and academic paper for submission

References

Muhammad, S., Abdulkareem, A., & Oyewusi, F. (2022). *NaijaSenti: A Nigerian Twitter Sentiment Corpus for NLP*. Proceedings of the Language Resources and Evaluation Conference (LREC).

Oyewusi, F., Mohammed, S., & Adetiba, E. (2020). *Enhancing VADER for Nigerian Pidgin Sentiment Analysis*. In Proceedings of the Workshop on African Language Technology.

Nekoto, W., Marivate, V., et al. (2020). *Participatory Research for Low-Resourced Machine Translation: A Case Study in African Languages*. Findings of EMNLP.

Agić, Ž., & Vulić, I. (2019). *JW300: A Wide-Coverage Parallel Corpus for Low-Resource Languages*. EMNLP-IJCNLP.

NLLB Team (2022). *No Language Left Behind: Scaling Human-Centered Machine Translation*. arXiv preprint arXiv:2207.04672.

Peer/Mentor Feedback

1. References Are Missing

- Although cited in-text, there's no formatted reference list. This weakens the credibility of the literature review.

2. Web App Concept Underdeveloped

- Mention of a demo app is promising but vague. What will the interface look like? What functionality will it prioritize?

3. Add More Specific Methodology Details

- While the expected outcomes are clear, the paper would benefit from a more detailed explanation of your methods. For instance, you mention collecting social media data and fine-tuning multilingual models, but it would help to briefly describe how data will be annotated, what platforms or scraping tools you'll use, or how you plan to evaluate model performance. This added specificity will strengthen the paper's credibility and make your approach easier to replicate or critique.

4. This proposal lacks a formal evaluation plan - while metrics like BLEU and F1 scores are mentioned in the timeline, there's no systematic methodology for validation or clear success thresholds. You need to define specific evaluation criteria, baseline comparisons, train/test splits, and performance benchmarks for both translation and sentiment analysis tasks. The timeline is well-structured, but should include more detailed evaluation milestones.

5. The work demonstrates a sophisticated understanding of both technical challenges and social implications, with a comprehensive approach that could make meaningful contributions to inclusive AI. The format is lacking.